

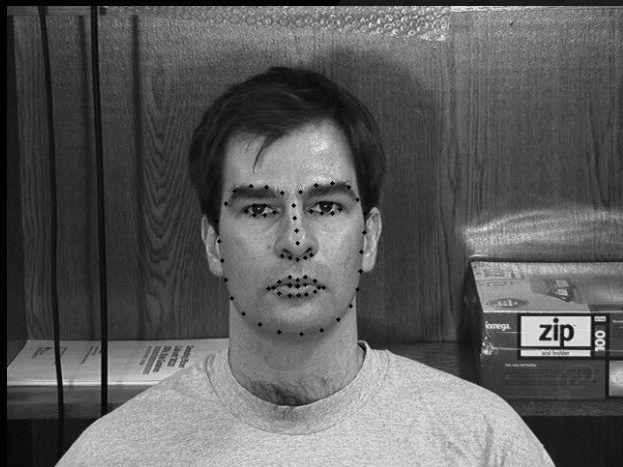


Masked & Unmasked Face Recognition

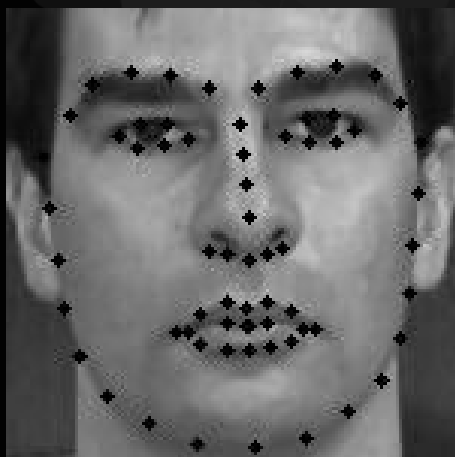
Group 11
July 24, 2023

Beginner Level

landmarks



crop and resize



prepare features
Hog

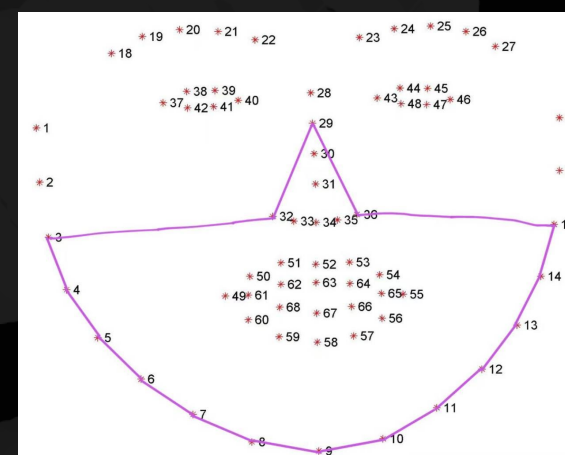
Binary classifier
SVM

Accuracy value
100%

digital mask added



key point selection

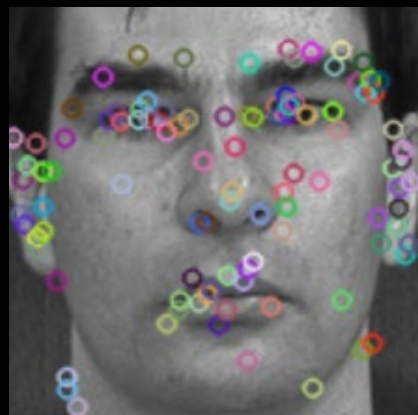


Expert Level

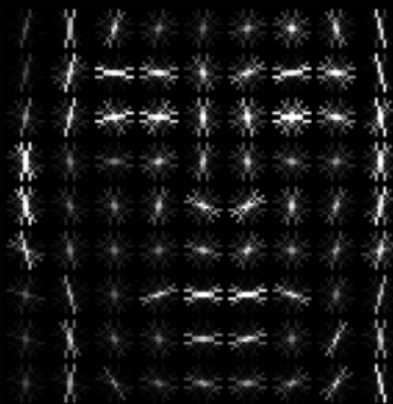
Why did we finally choose to extract Sift features?



Sift



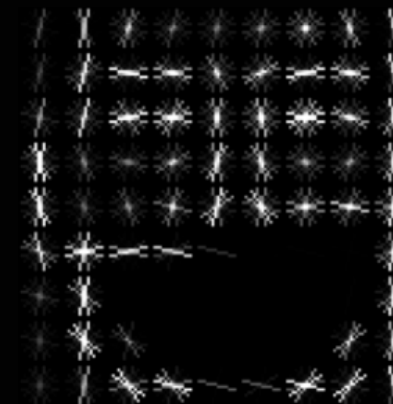
Hog



Sift



Hog



Expert Level — different pre-processing



Pre-Processing Plus



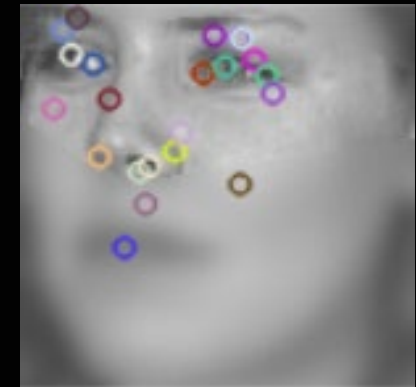
Minimum value filter



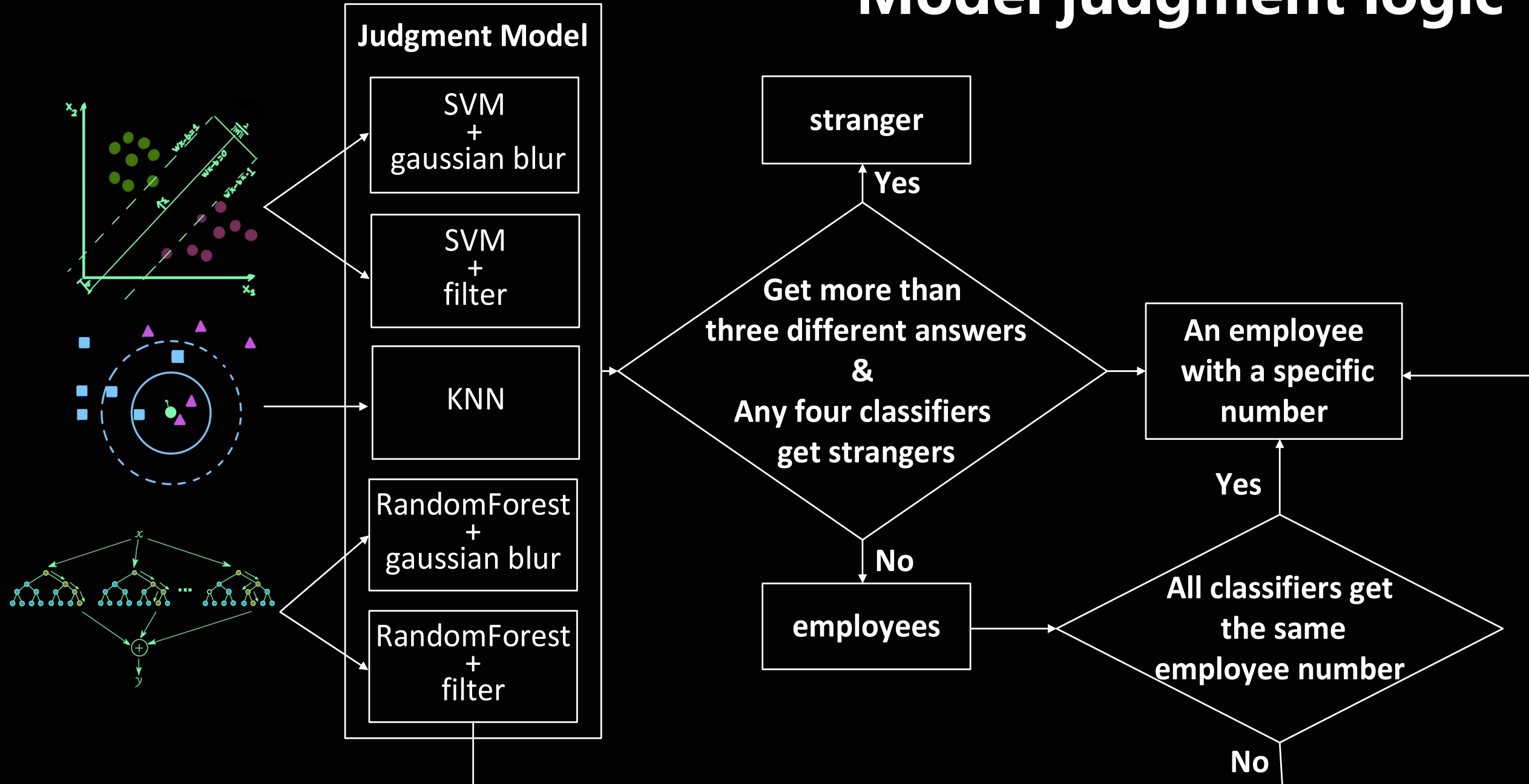
Enhance Brightness

Redefine the treatment area

Gaussian blur



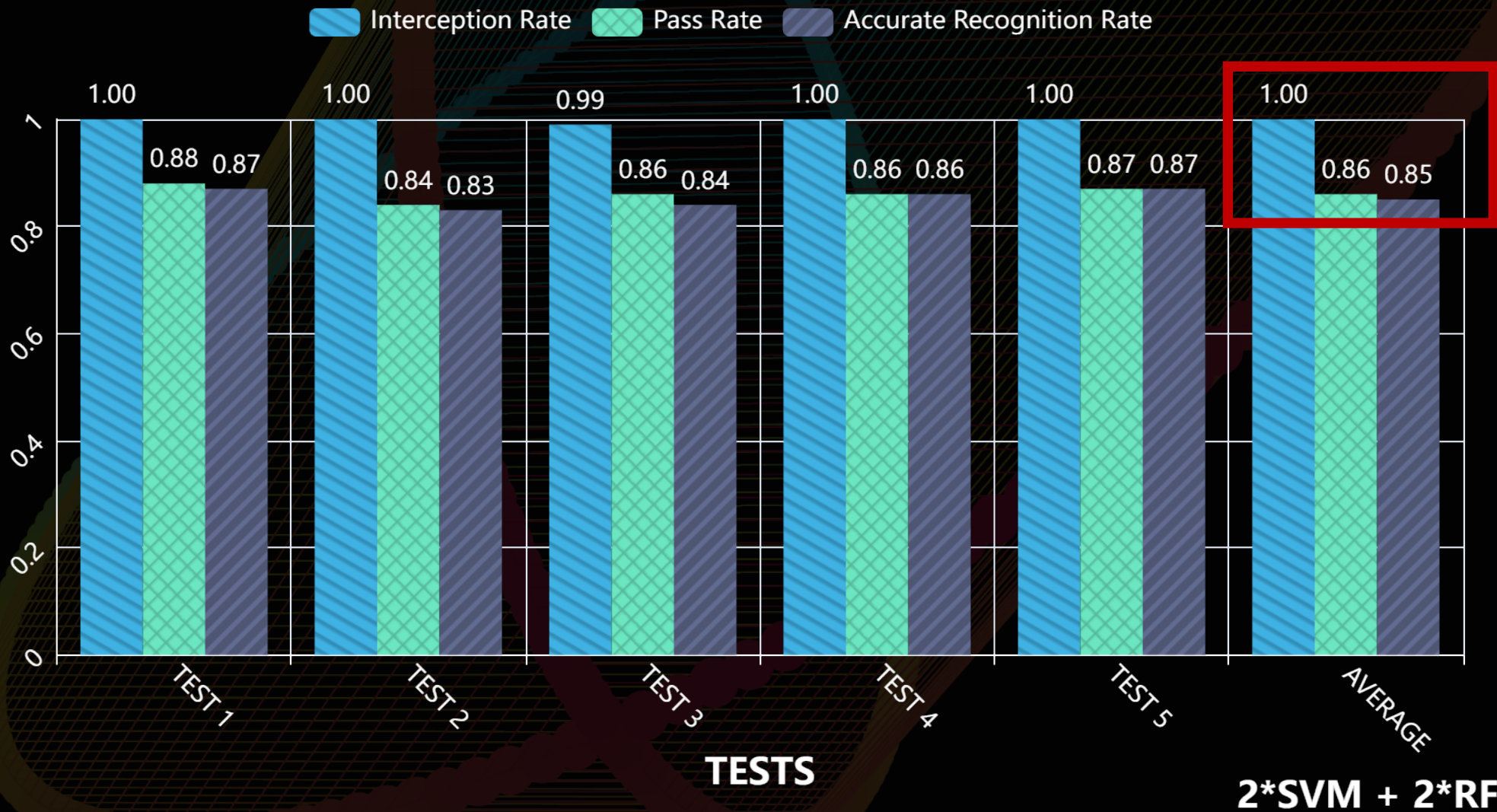
Model judgment logic



Unmasked Face Recognition

SVM

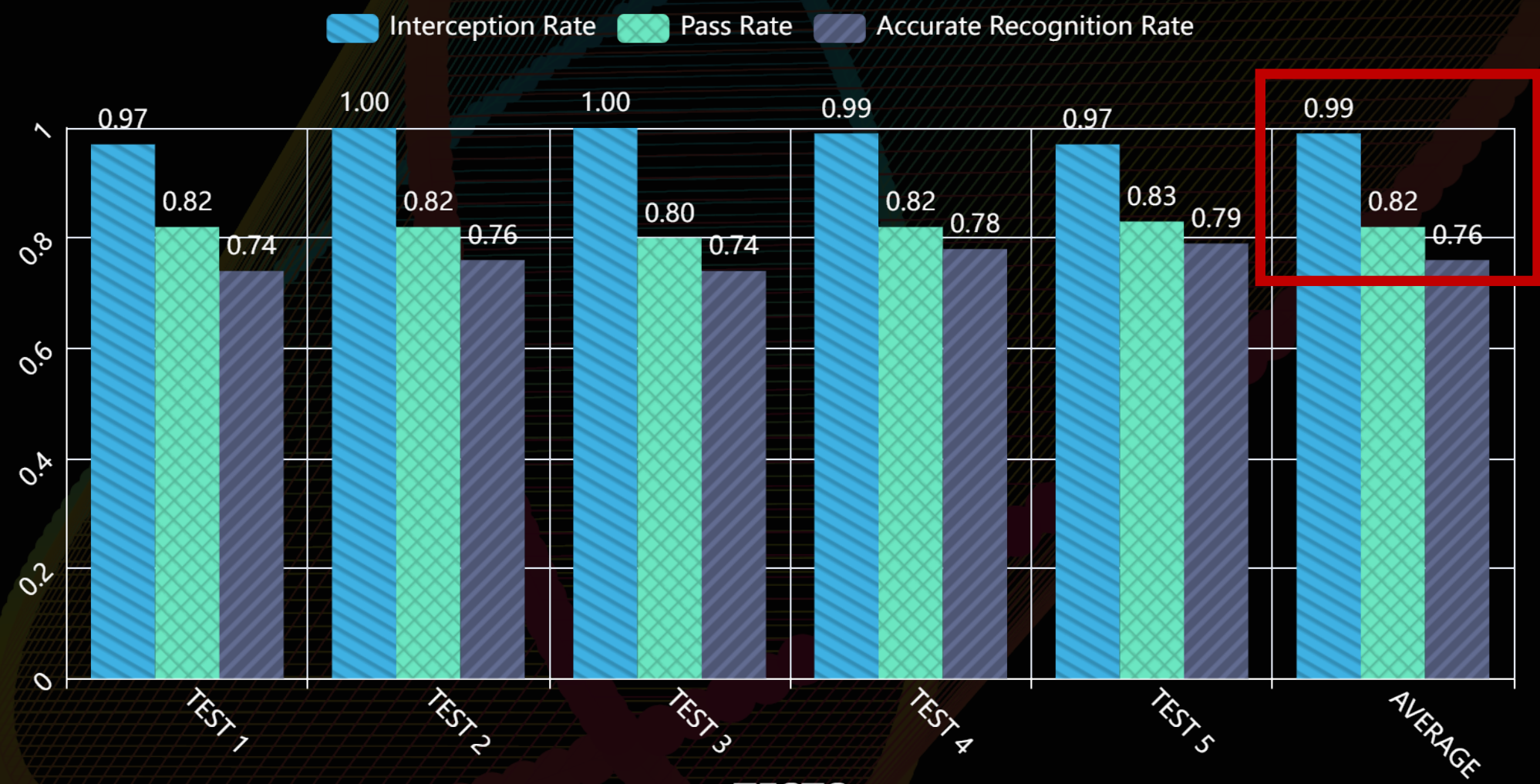
Interception Rate : 0.81
Pass Rate : 0.75
Accurate Recognition Rate : 0.74



masked Face Recognition

SVM

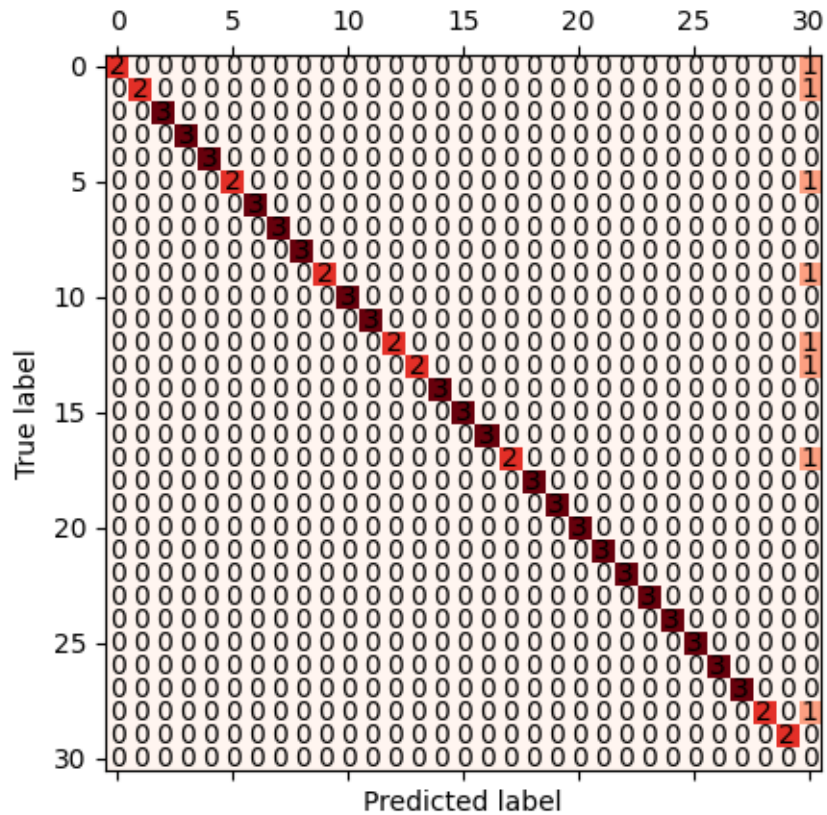
Interception Rate : 0.72
Pass Rate : 0.59
Accurate Recognition Rate : 0.57



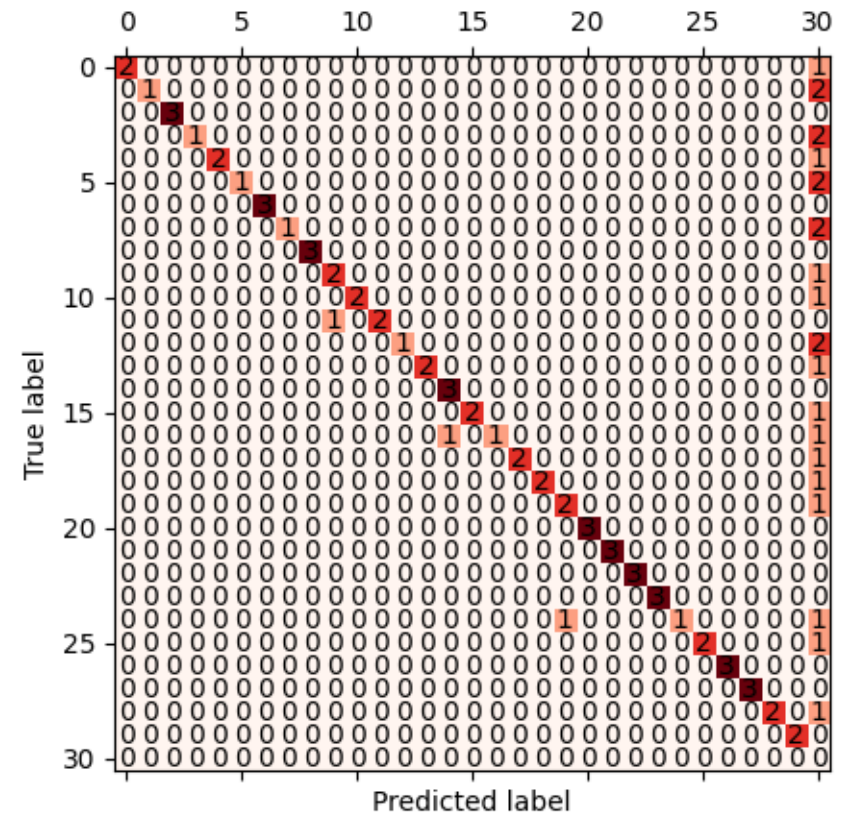
TESTS

2*SVM + 2*RF + KNN

Confusion Matrix

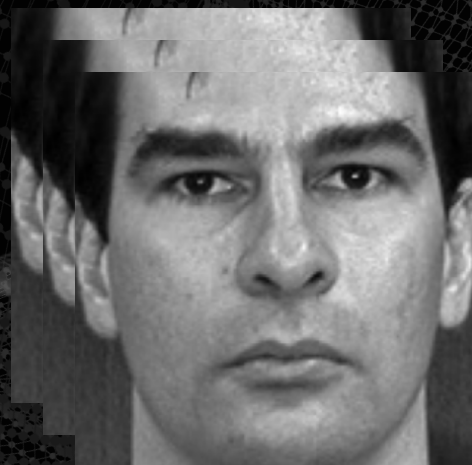


Unmasked

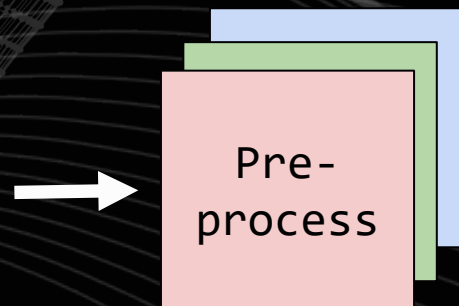


Masked

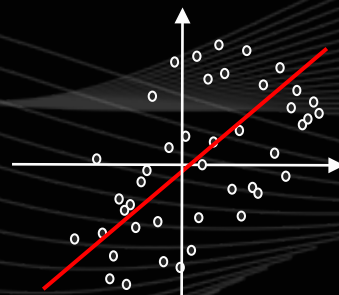
Bonus Level ---Georgia Tech Faces



Training samples



fitting

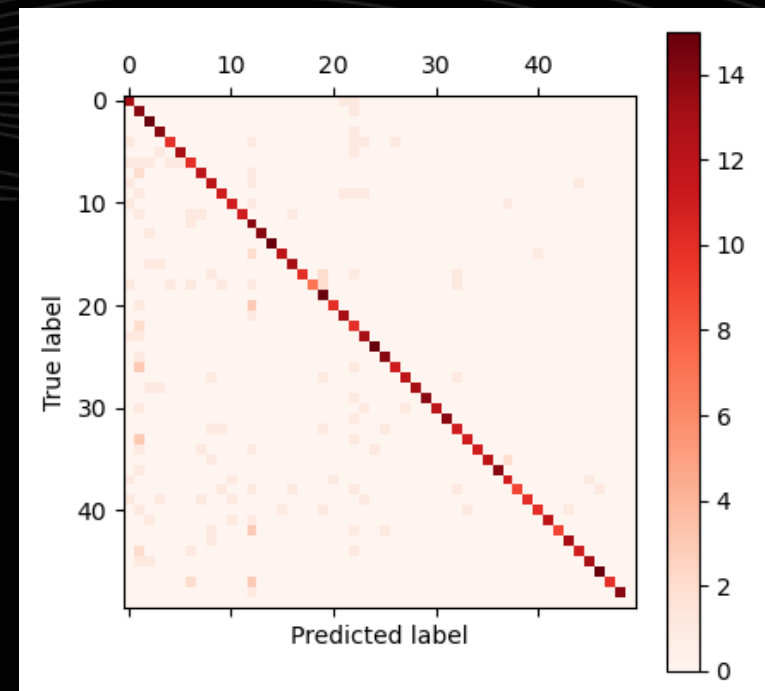
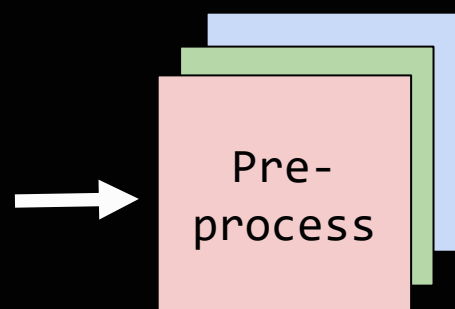


SVM model

predict



Testing samples

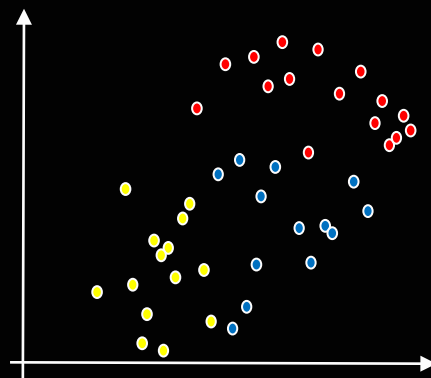


Accuracy: 80.8%

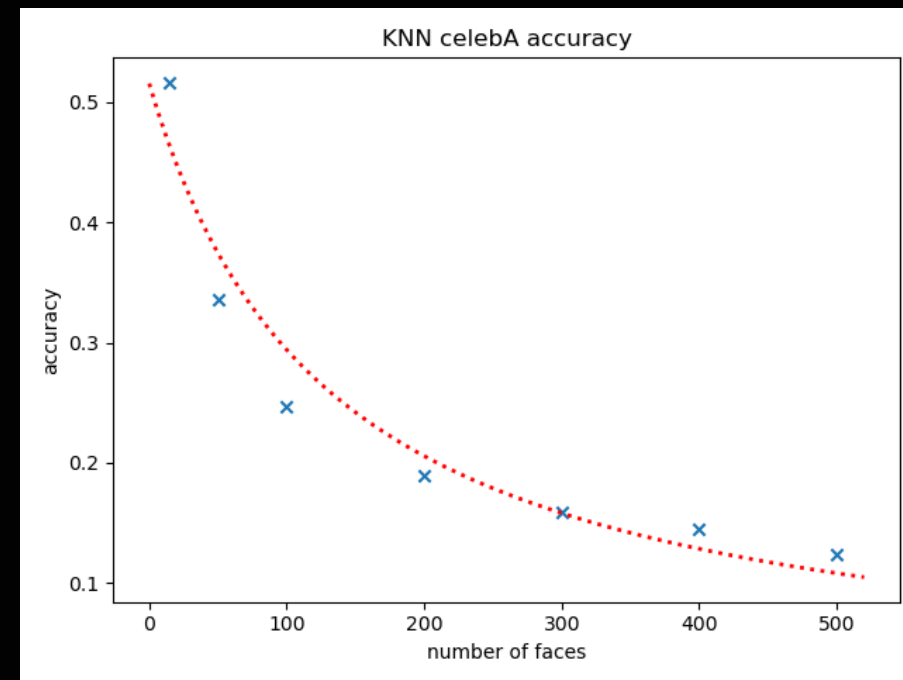
Bonus Level ---celeb A



Large-scale dataset



KNN model



Poor data quality



Dataset
cleaning

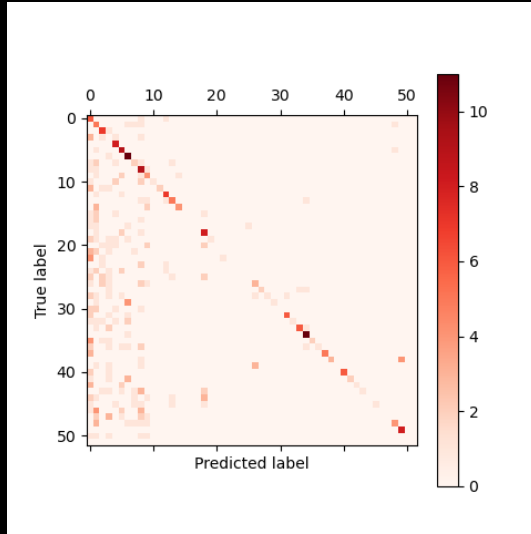
500 class accuracy:
14.15% (1.75% improved)



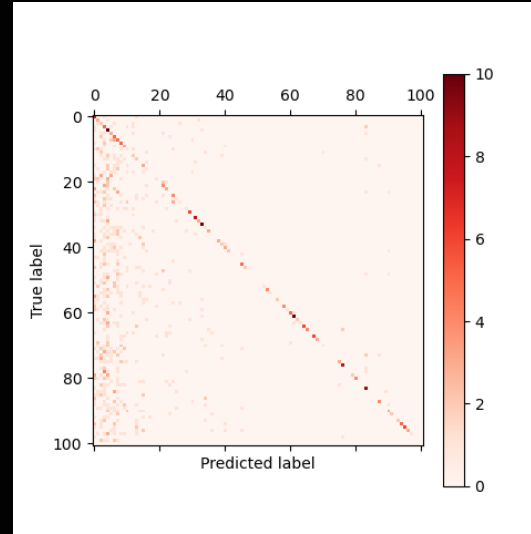
Q&A

- 后面还可以加一些补充材料

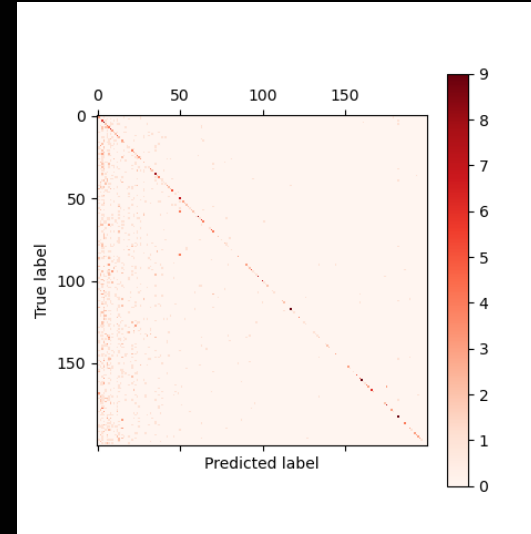
Supplementary materials



Bonus-50
Acc-33.61%

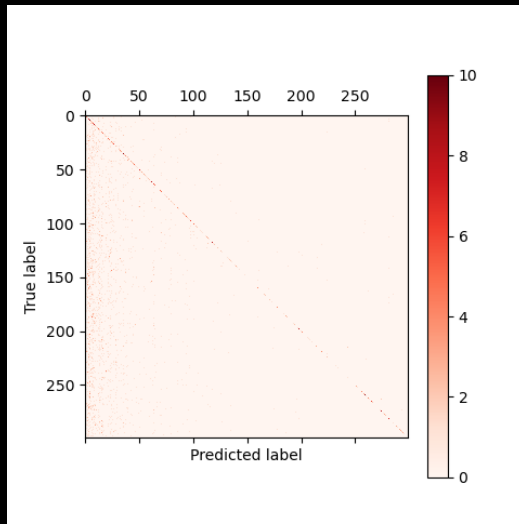


Bonus-100
Acc-24.68%

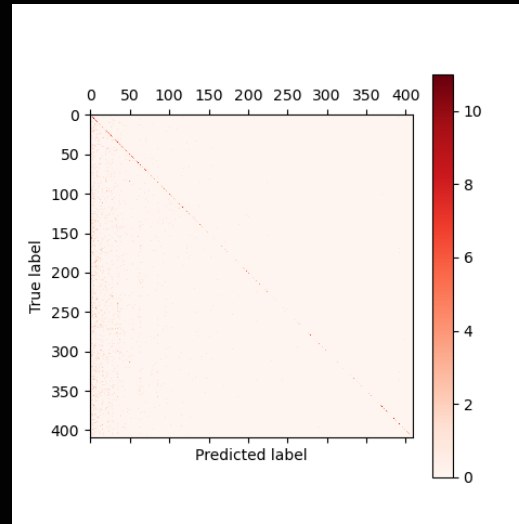


Bonus-200
Acc-18.90%

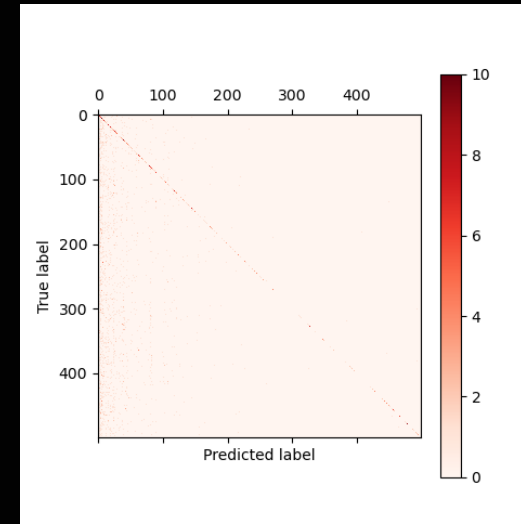
Supplementary materials



Bonus-300
Acc-15.86%



Bonus-400
Acc-14.45%



Bonus-500
Acc-14.15%